

BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: BIT/2024/43959

School/Faculty: COMPUTING AND INFORMATICS

Programme: BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Academic Year: YEAR 4

Semester: 1

Lecturer: MR MICHAEL NYORO

Phone Number: 0796800979

Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

only screenshots needed and labeling

Screenshots/Evidence Attached (Yes/No)

Tools/Software Used

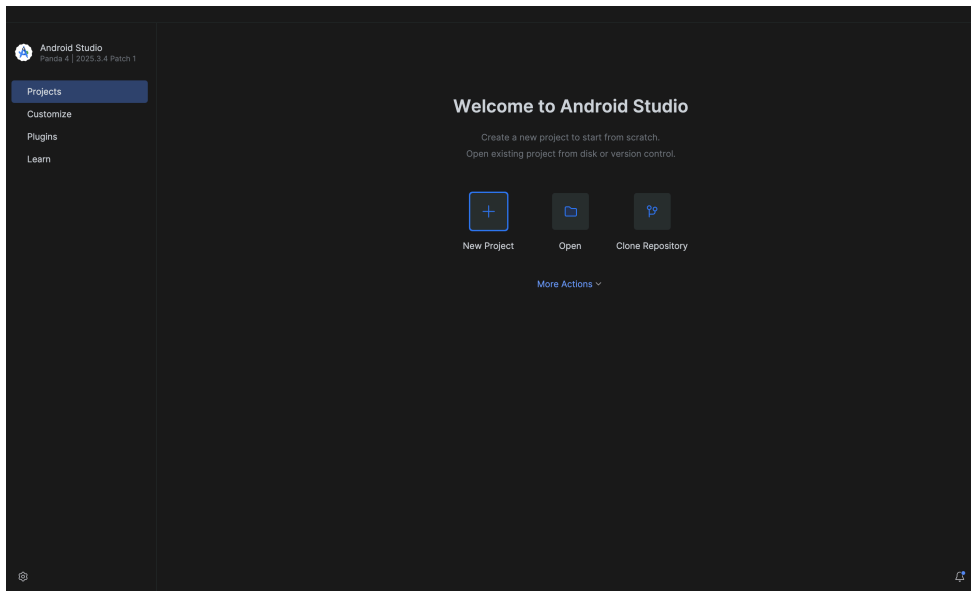


Fig 1.1: Android studio installation

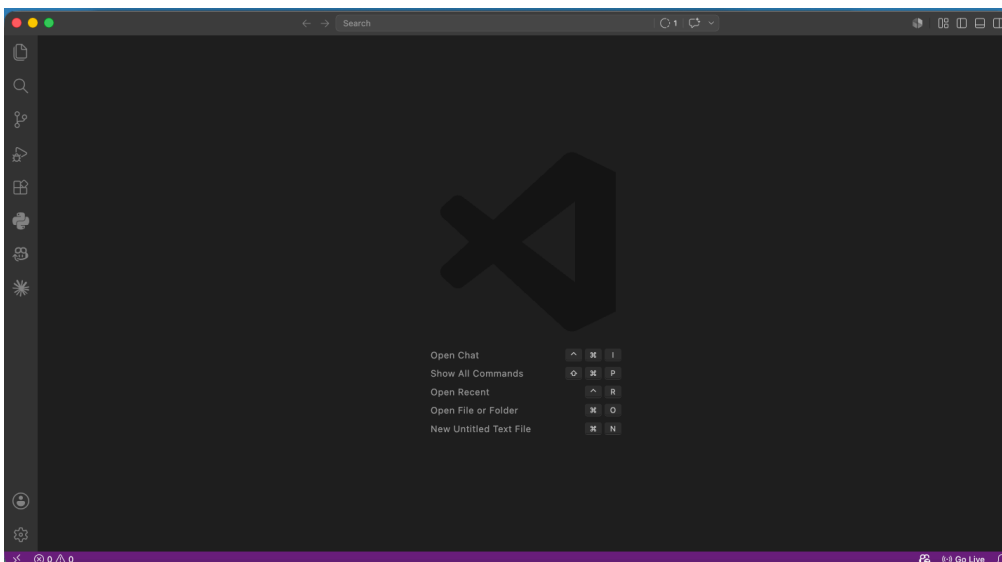


Fig 1.2: VSCode installation

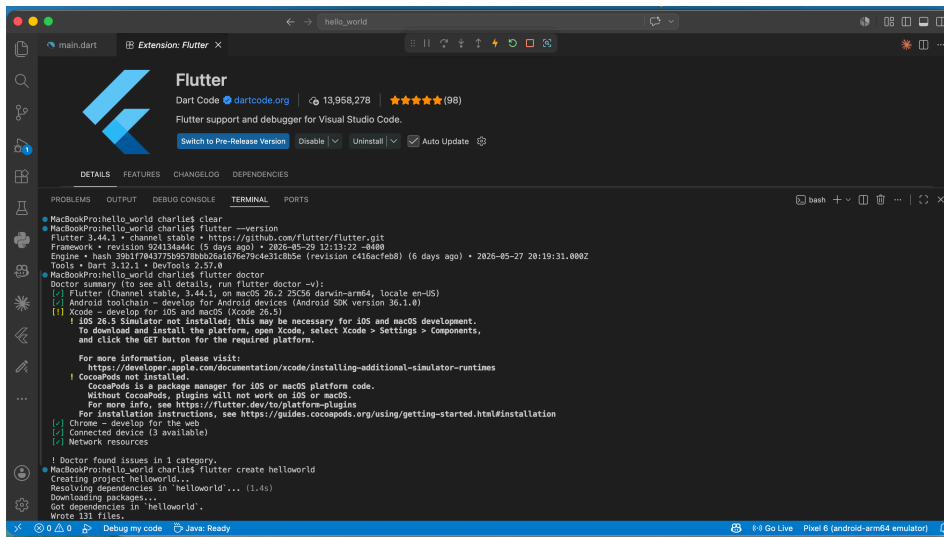


Fig 1.3: Flutter and Flutter extension installation

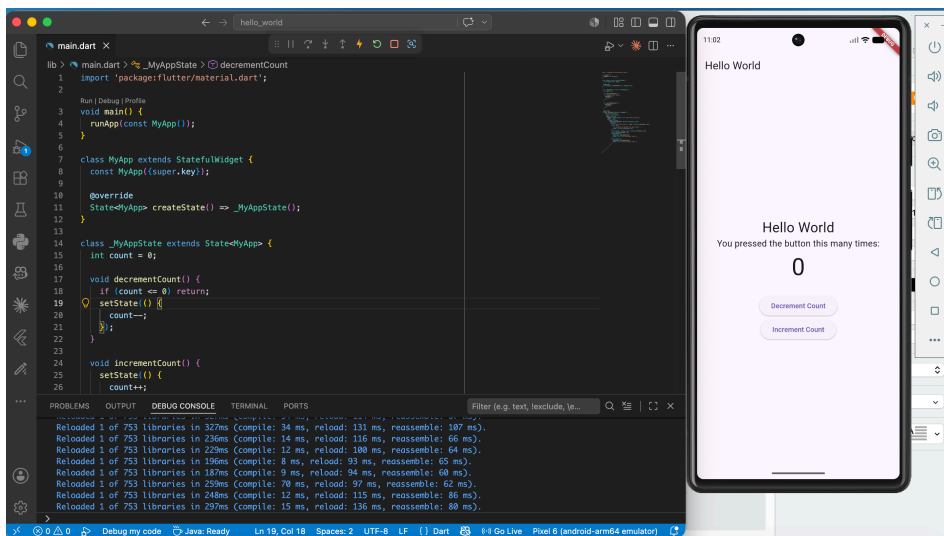


Fig 1.4: Hello world counter app

Personal Reflection

Flutter/dart appears to have a potentially good learning curve based on previous programming knowledge.

Week 2: Mobile Development Environment

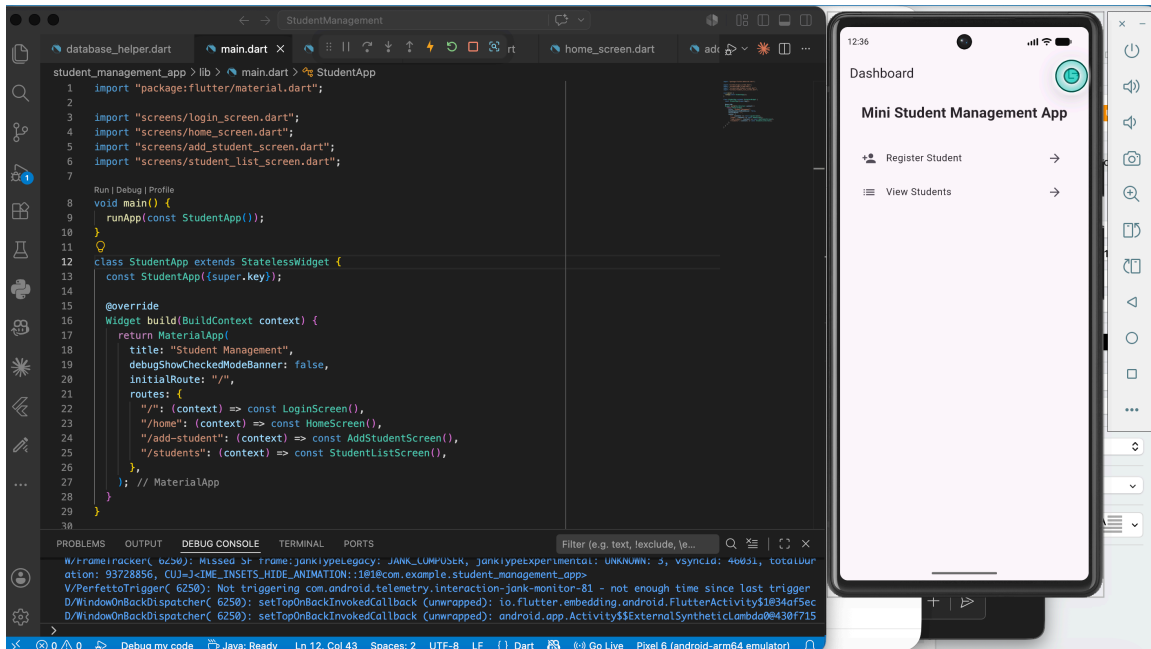


Fig 2.1: Mini student management app

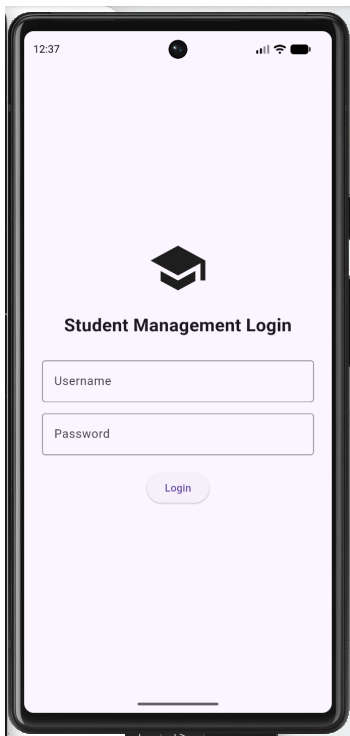


Fig 2.2: Admin Login page

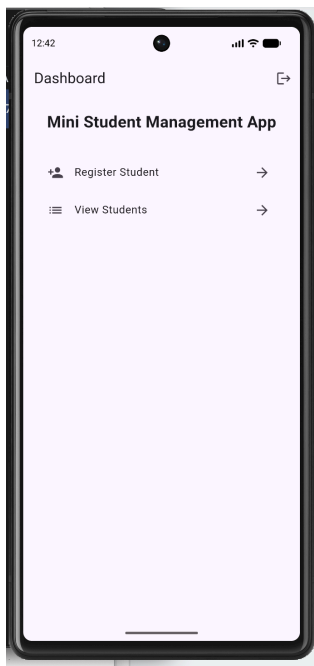


Fig 2.3: Homepage

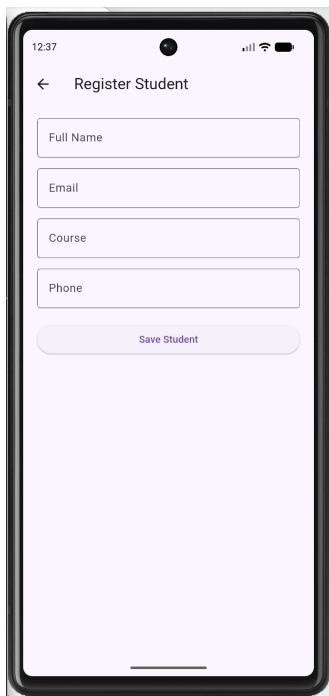


Fig 2.4: Register student

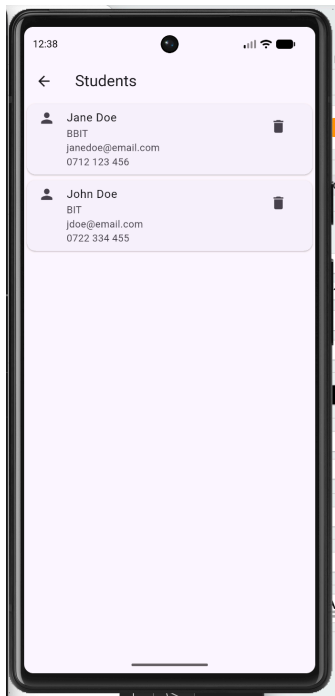


Fig 2.5: List students

Personal Reflection

Interesting insights on persisting data permanently on a local database.

Week 3: User Interface Design

Same as week 1

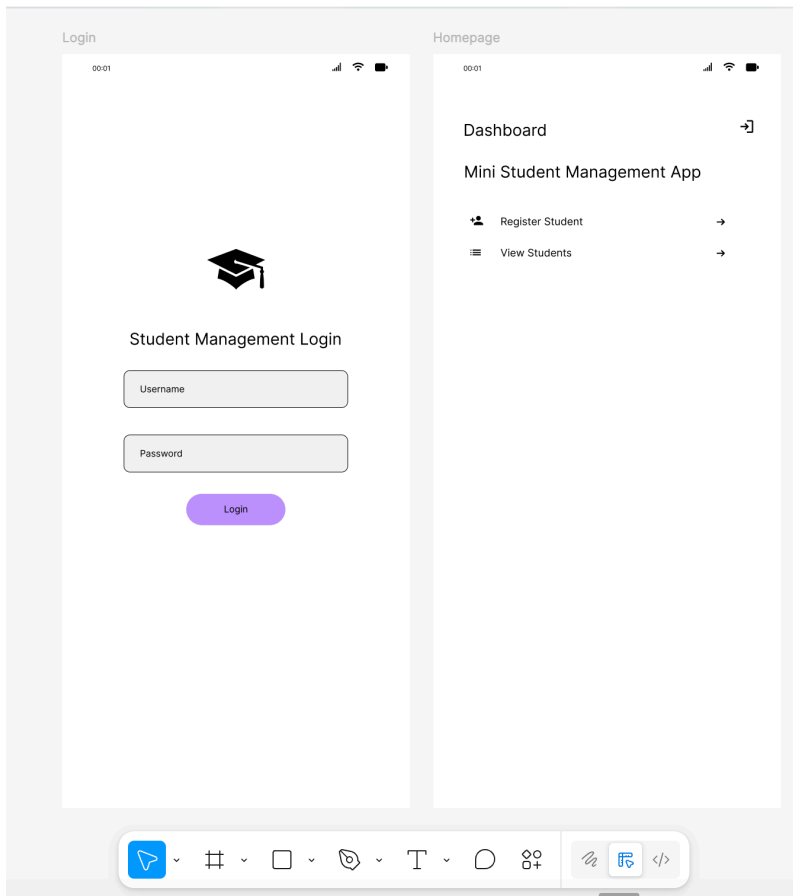


Fig 3.1: Figma - Login and Homepage wireframes

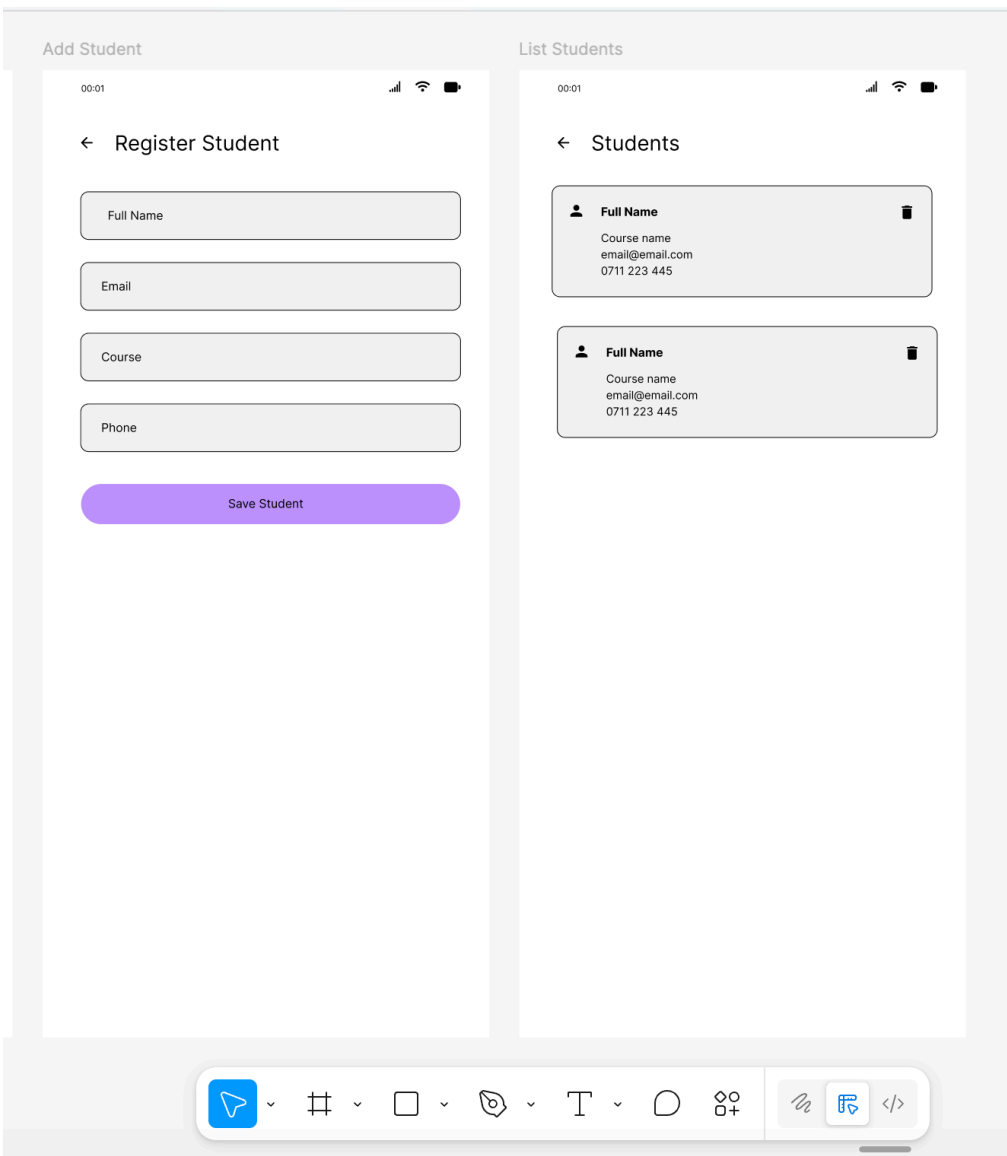


Fig 3.2: Figma - Register and List Students wireframes

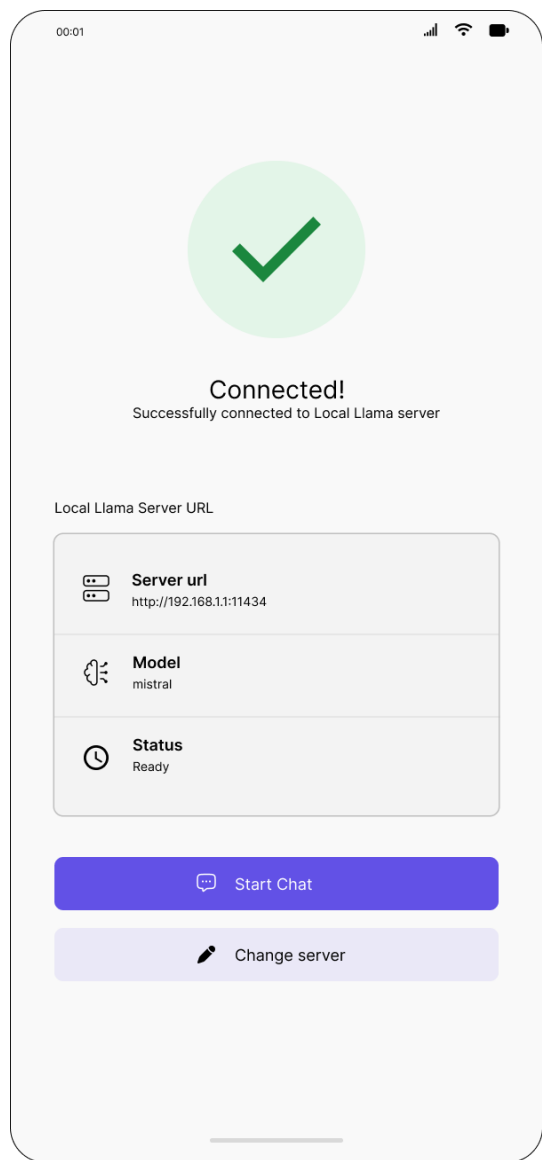
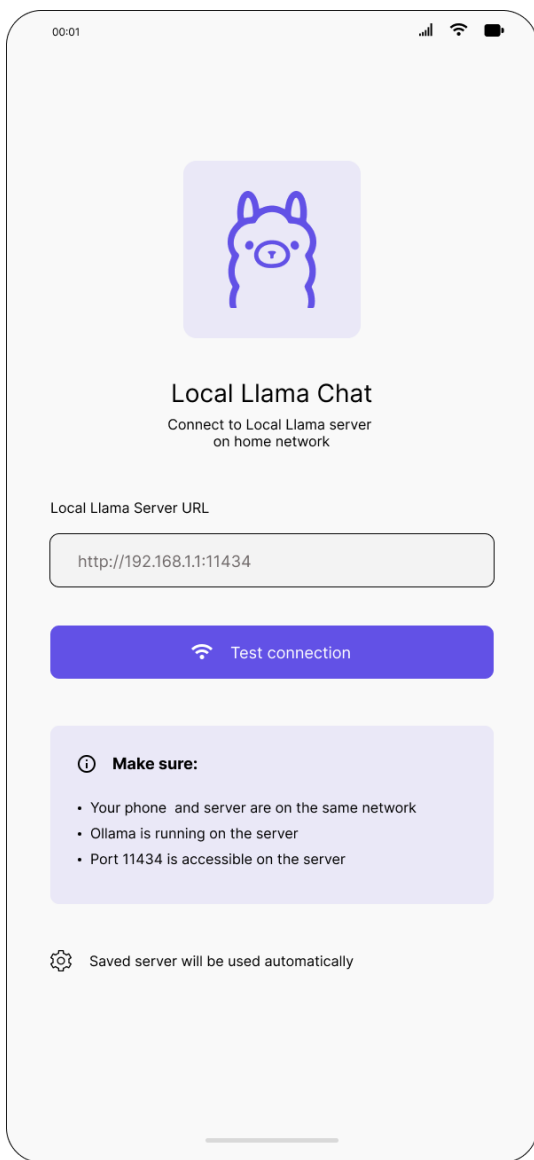


Fig 3.3: Figma - Local Llama Homepage Wireframe

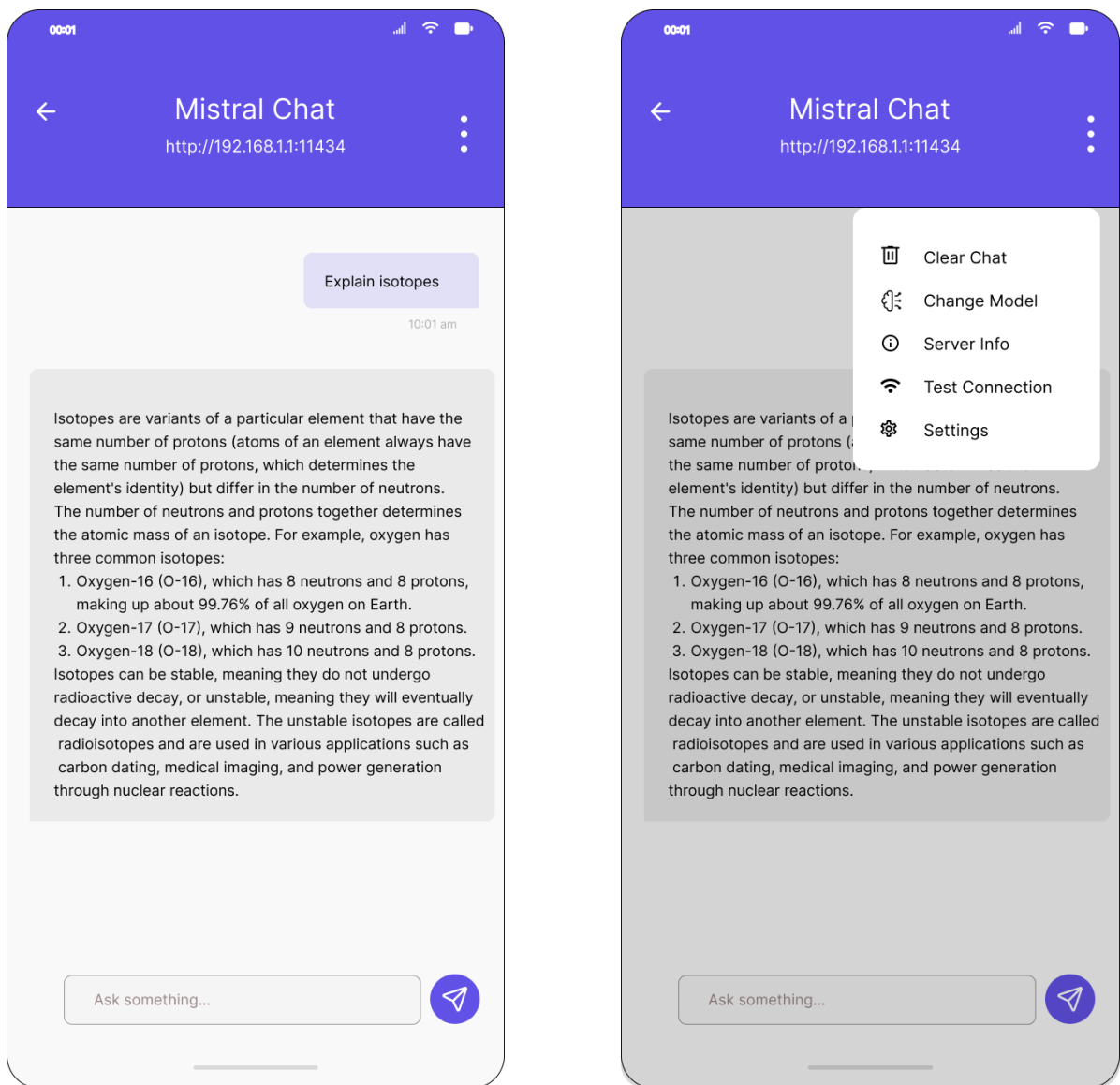


Fig 3.3: Figma - Local Llama Chat Wireframes

Personal Reflection

Prototyping first will speed up development.

Week 4: Event Handling and User Interaction

Same as week 1

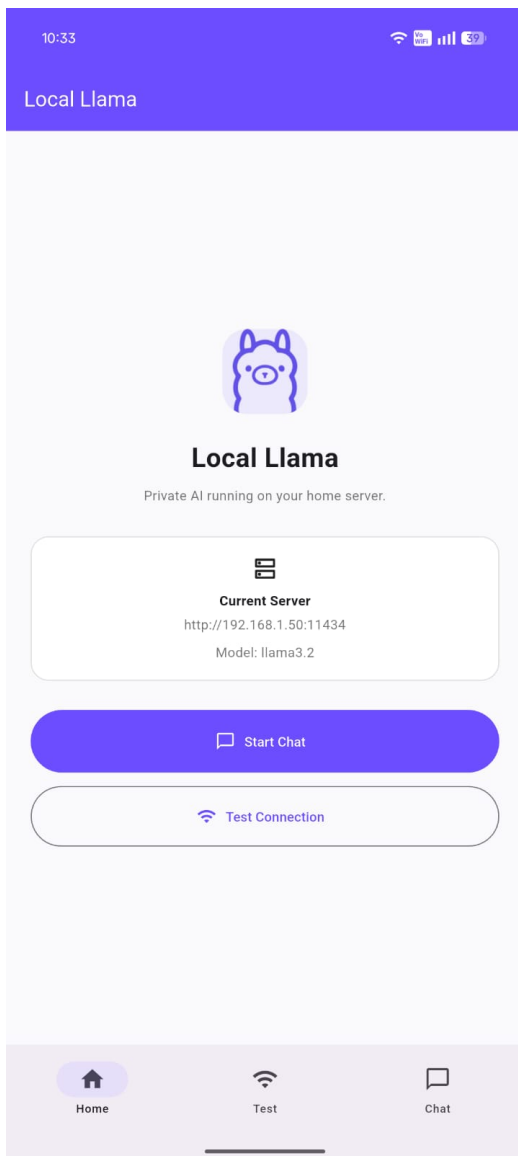


Fig 3.3:

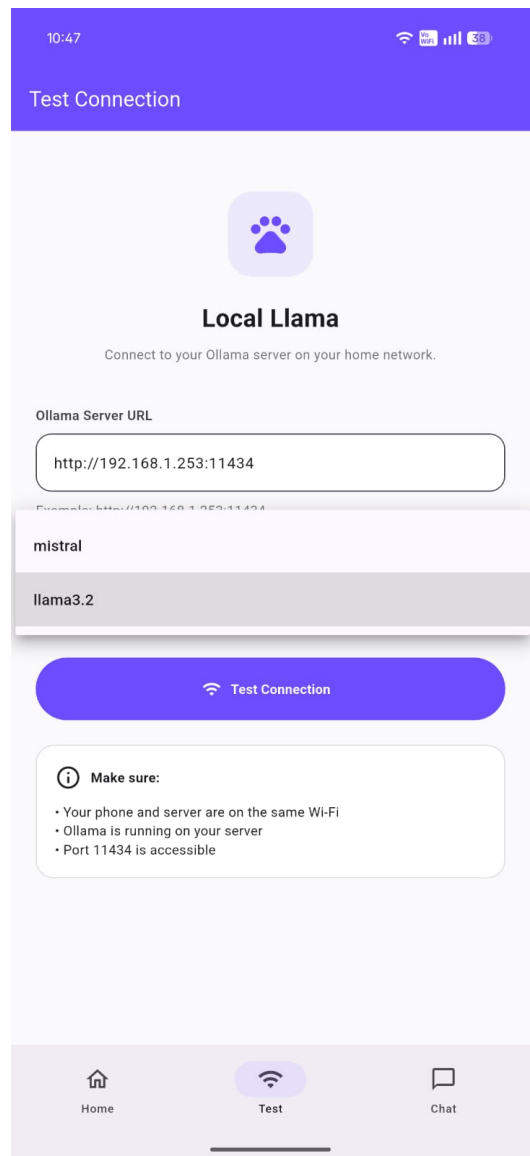


Fig 4.1: Live app - Intuitive navigation

Fig 4.2: AI Model selection

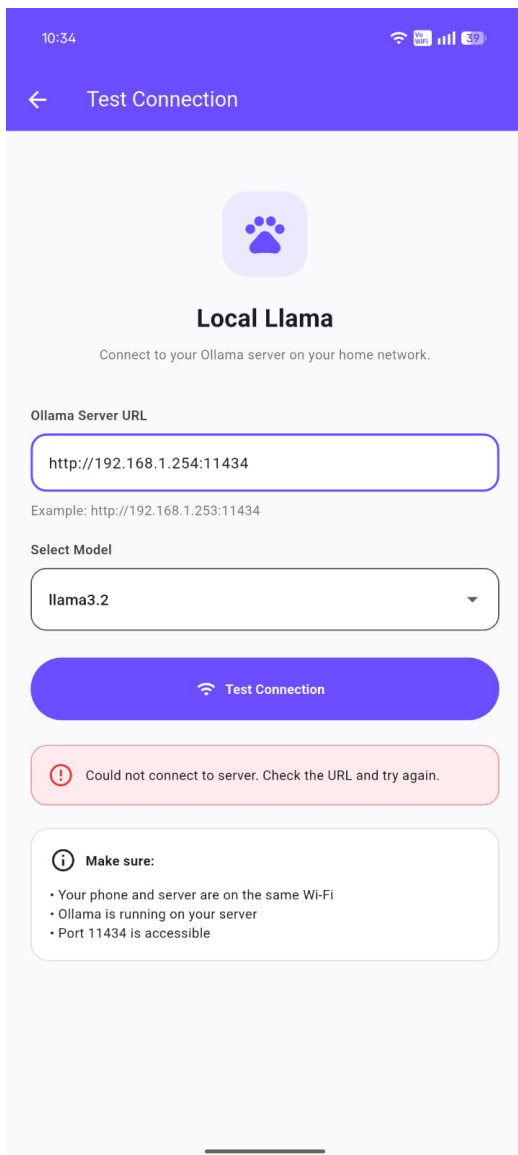


Fig 4.3: Connection event failure

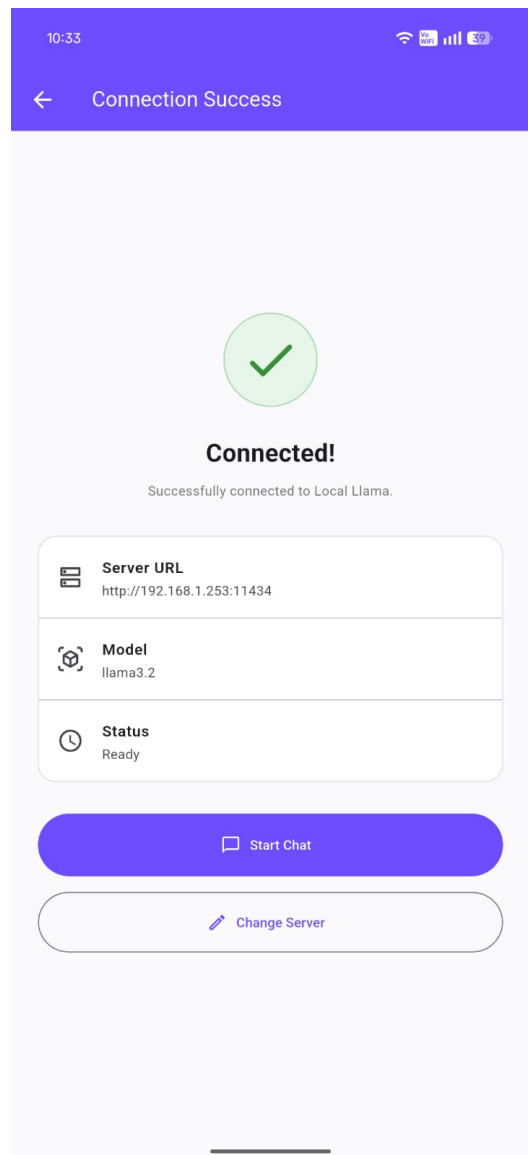


Fig 4.4: Connection event success

Week 5: Data Management

Same as week 1

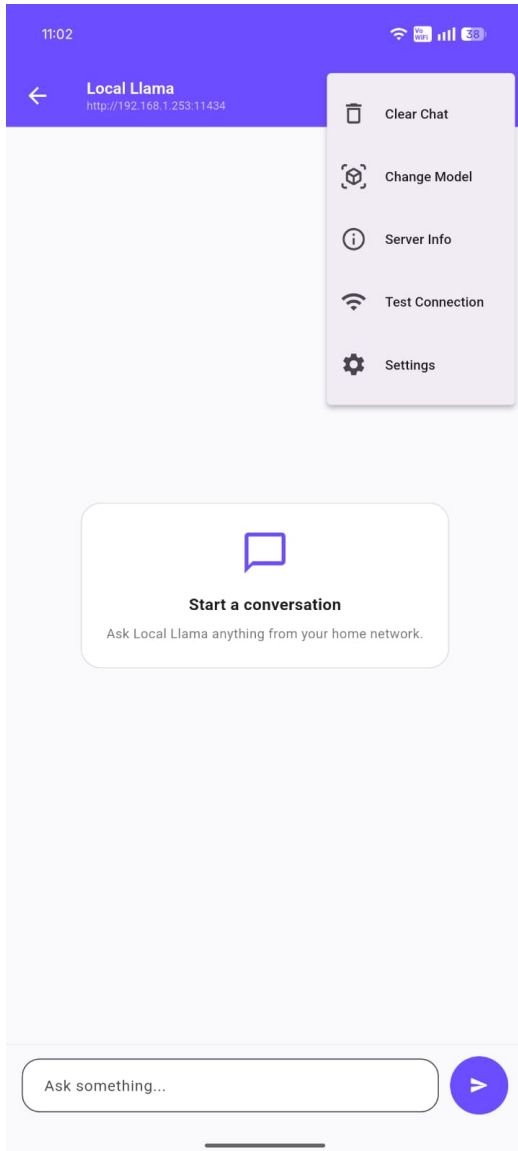


Fig 5.1: App settings data

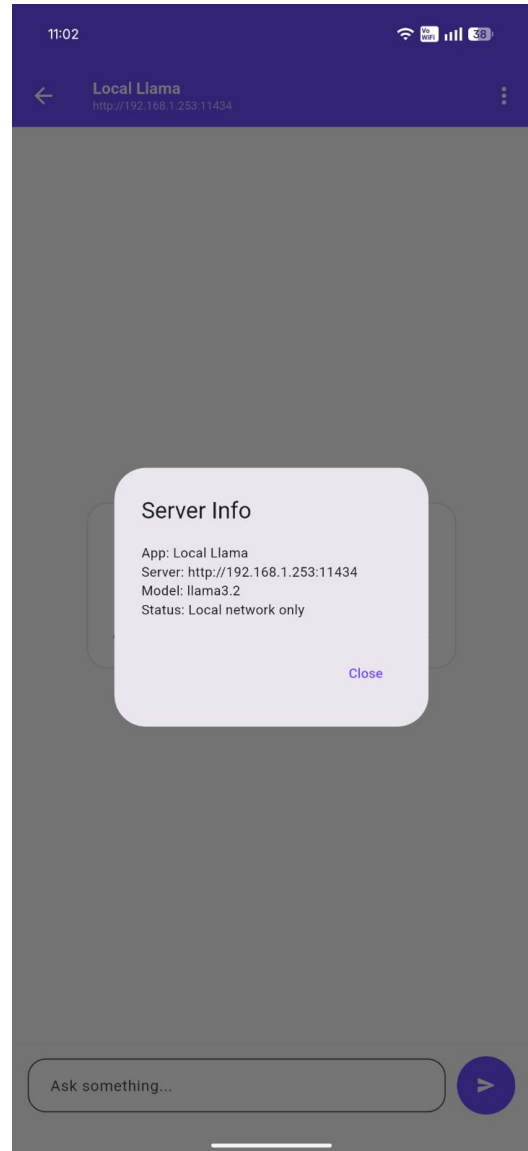


Fig 5.2: Data in shared preferences

Week 6: Networking and APIs

Same as week 1

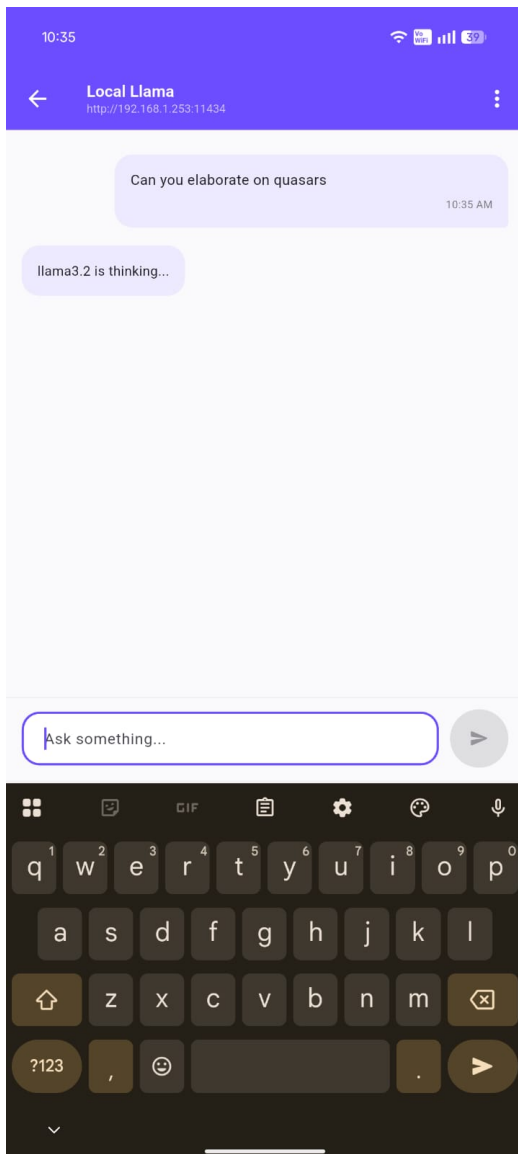


Fig 6.1: POST data to server

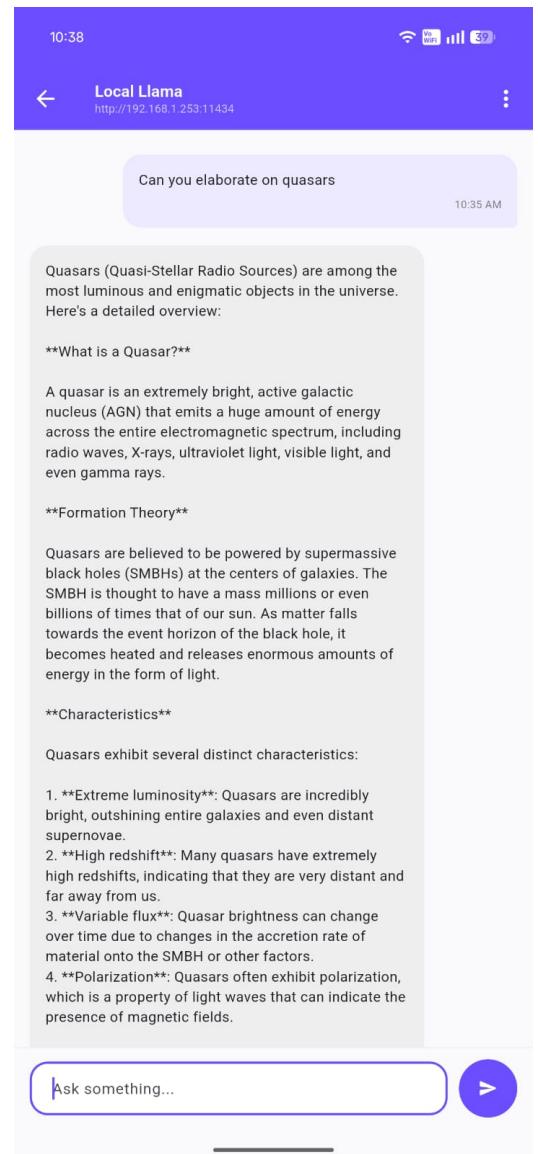


Fig 6.2: Data received from server

Week 7: Application Architecture

Same as week 1

Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: _____

Student Signature: _____

Date: _____

Lecturer Name: _____

Lecturer Remarks: _____

Date: _____